

TECH & AI REPORT 2025

The Definitive Guide to Artificial Intelligence,
Emerging Technologies & the Digital Future

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\$2.7T

Global AI Market by 2032

73%

Enterprises Adopting Generative
AI

300M+

Jobs Impacted by Automation

40%

Annual Growth Rate (AI Sector)

TOPICS COVERED IN THIS REPORT

Generative AI | Large Language Models | Cloud Computing | Quantum Computing
Cybersecurity | Edge AI | Autonomous Systems | AI Ethics & Regulation

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1. Executive Summary

1.1 Key Findings

The year 2025 marks an inflection point in the history of technology. Artificial Intelligence has transitioned from a specialised research discipline into the foundational layer of the global economy. This report synthesises data from over 500 industry sources, surveys of 10,000+ enterprise leaders, and the latest academic research to present a comprehensive view of the technological landscape.

Key Insight: For the first time, AI-related capital expenditure surpassed traditional IT spending in Q1 2025, signalling a fundamental reordering of enterprise technology priorities.

Top findings from this year's research:

- Generative AI adoption in enterprises grew 340% year-on-year, with 73% of Fortune 500 companies deploying production AI systems.
- The global AI market is projected to reach \$2.7 trillion by 2032, growing at a CAGR of 40.2%.
- Edge AI deployments tripled in 2024, driven by demand for real-time inference without cloud dependency.
- AI cybersecurity tools detected and neutralised threats 8x faster than traditional rule-based systems.
- Quantum computing achieved commercial viability milestones in pharmaceutical discovery and logistics optimisation.
- 67% of knowledge workers now use AI assistants daily, up from 18% in 2023.
- The EU AI Act came into force, establishing the world's first comprehensive AI regulatory framework.

1.2 Report Scope

This report covers developments across the global technology sector from January 2024 to May 2025. Data points are drawn from primary research, vendor disclosures, regulatory filings, and peer-reviewed publications. Geographic coverage spans North America, Europe, Asia-Pacific, and emerging markets in Africa and Latin America. Unless stated otherwise, all financial figures are in USD.

2. State of AI in 2025

2.1 Market Overview

Artificial Intelligence is no longer an emerging technology — it is the engine of modern commerce. From healthcare diagnostics to supply chain optimisation, AI permeates every sector of the economy. The following table summarises the AI market by segment:

AI Segment	2024 Market Size	2025 Estimate	CAGR (5yr)
Generative AI	\$67B	\$105B	48.3%
Machine Learning Platforms	\$41B	\$58B	22.1%
AI Chips & Hardware	\$88B	\$134B	35.7%
AI-as-a-Service (AlaaS)	\$29B	\$47B	31.4%
Robotics & Automation	\$53B	\$71B	18.9%
Computer Vision	\$19B	\$27B	24.6%
Natural Language Processing	\$25B	\$38B	29.8%

Source: Tech & AI Insights Market Analysis, Q1 2025

2.2 Leading AI Companies

A cohort of hyperscalers and AI-native companies dominate the landscape, collectively accounting for over 60% of global AI R&D spend:

Company	HQ	AI Focus Area	AI Revenue (2024)
Microsoft / OpenAI	USA	LLMs, Copilot, Azure AI	\$38.2B
Google / DeepMind	USA	Gemini, Search AI, TPUs	\$34.7B
Amazon Web Services	USA	Bedrock, SageMaker, Alexa	\$29.1B
Meta AI	USA	LLaMA, Social AI, AR/VR	\$18.4B
Anthropic	USA	Claude, Constitutional AI	\$6.8B
Baidu	China	ERNIE, Autonomous Driving	\$9.2B
Nvidia	USA	AI Hardware (GPUs, NIM)	\$44.9B
IBM	USA	Watson X, Enterprise AI	\$5.6B

2.3 Funding Landscape

Venture capital investment in AI reached an all-time high of \$131 billion in 2024, representing 38% of all global VC deployment. Early-stage AI deals increased by 210% while late-stage rounds averaged \$450 million — reflecting investor confidence in AI's long-term commercial potential.

- Top funded category: Foundation Models & LLMs (\$41B, 31% of total AI VC)
- Runner-up: AI Infrastructure & Hardware (\$28B)

- Fastest growing: AI for Healthcare (\$17B, +180% YoY)
- Geographic leader: USA (54%), followed by China (19%), Europe (15%)

3. Generative AI Revolution

3.1 Large Language Models (LLMs)

Large Language Models represent the most transformative AI development of the decade. These neural networks, trained on trillions of tokens of text data, exhibit remarkable capabilities: writing code, composing legal documents, analysing scientific literature, and engaging in complex multi-turn reasoning. The model landscape has matured rapidly:

Model	Developer	Parameters	Context Window	Key Capability
GPT-4o	OpenAI	~200B	128K tokens	Multimodal reasoning
Claude 3.7 Sonnet	Anthropic	Undisclosed	200K tokens	Extended thinking
Gemini 2.0 Ultra	Google	~1.5T (MoE)	1M tokens	Long context mastery
LLaMA 4	Meta	405B	128K tokens	Open-source frontier
Mistral Large 3	Mistral AI	~70B	64K tokens	Efficient reasoning
DeepSeek R2	DeepSeek	671B (MoE)	64K tokens	STEM & coding

3.2 Multimodal AI

The convergence of language, vision, audio, and action into unified models has unlocked entirely new application categories. Multimodal AI systems can now interpret medical scans while explaining findings in plain language, generate product videos from text briefs, and control software applications through visual perception.

Breakthrough 2025: Google's Gemini 2.0 demonstrated native audio understanding and generation, real-time video analysis, and code execution in a single unified model — blurring the line between AI assistant and AI agent.

3.3 AI-Generated Content

Generative AI has permanently altered the content creation industry. By 2025, an estimated 15% of all digital content published globally involved AI generation or AI assistance. This spans text, images, video, music, and code:

- Text: 42% of marketing copy now drafted with AI assistance (CMO Survey, 2025)
- Code: GitHub Copilot and similar tools contribute to 55% of all public commits on major platforms
- Images: Adobe Firefly, Midjourney, and DALL-E collectively generate 34 billion images per month
- Video: AI video generation platforms processed 8.2 million hours of synthetic video in Q4 2024
- Music: AI music platforms serve 90 million monthly listeners with AI-composed or AI-assisted tracks

4. AI in Industry

4.1 Healthcare & Life Sciences

Healthcare represents one of AI's highest-impact application domains. AI systems now assist with everything from diagnostic imaging to drug discovery, clinical trial design, and patient monitoring. The results are measurable and compelling:

- AI-assisted pathology reduced diagnostic errors by 32% at leading hospital networks (NEJM, 2024)
- AlphaFold 3 predicted protein-ligand interactions, accelerating drug discovery pipelines by 70%
- AI triage systems reduced ER wait times by 28% across 340 hospitals in pilot programmes
- Surgical robotics powered by AI vision achieved precision improvements of 4x over human-only procedures
- Mental health chatbots provided support to 140 million users globally in 2024

4.2 Financial Services

The financial sector is among the earliest and deepest adopters of AI, leveraging it for fraud detection, algorithmic trading, credit scoring, regulatory compliance, and personalised wealth management. Key metrics from 2024:

Use Case	Adoption Rate	Avg. ROI	Leader
Fraud Detection	89%	620%	Mastercard Decision AI
Algorithmic Trading	74%	340%	Jane Street, Citadel
Credit Scoring	71%	180%	Upstart, ZestAI
Customer Service AI	83%	210%	JPMorgan Chase, HSBC
Regulatory Compliance	62%	150%	ComplyAdvantage
Risk Management	58%	190%	Palantir Foundry

4.3 Manufacturing & Industry 4.0

Smart manufacturing, powered by AI, IoT, and robotics, is delivering unprecedented gains in productivity and quality. Predictive maintenance alone saves the global manufacturing sector an estimated \$630 billion annually by preventing unplanned downtime.

- Computer vision quality inspection systems achieve 99.98% defect detection accuracy
- Digital twin technology, augmented by AI, reduces prototype-to-production time by 45%
- AI-optimised supply chains recovered 3x faster from disruptions versus traditional approaches
- Collaborative robots (cobots) with AI vision work safely alongside humans in 28,000+ factories

4.4 Education & EdTech

AI is fundamentally reshaping pedagogy. Personalised learning systems adapt content difficulty and style to each student's pace, while AI tutors provide 24/7 subject support. Early results from

longitudinal studies show 18–34% improvements in learning outcomes when AI tools are thoughtfully integrated into curricula.

5. Cloud Computing & Infrastructure

5.1 Hyperscalers & AI Cloud Services

The three major hyperscalers — AWS, Microsoft Azure, and Google Cloud — are engaged in an unprecedented infrastructure arms race. Combined capital expenditure on AI-focused data centre expansion exceeded \$230 billion in 2024, with projections of \$380 billion for 2025. AI workloads now represent 38% of all cloud compute consumption.

Cloud Provider	2024 Revenue	Market Share	AI Flagship Product
AWS	\$105B	31%	Amazon Bedrock, SageMaker
Microsoft Azure	\$93B	27%	Azure OpenAI Service
Google Cloud	\$54B	16%	Vertex AI, Gemini API
Alibaba Cloud	\$17B	5%	Tongyi Qianwen
Oracle Cloud	\$12B	4%	OCI AI Services
IBM Cloud	\$8B	2%	Watson X Platform
Others	\$51B	15%	Various

5.2 AI-Optimised Hardware

The AI hardware revolution is reshaping semiconductor design. Nvidia's GPU dominance continues with the H200 and Blackwell B200 architectures, but challengers are emerging from all directions:

- Nvidia H200 / B200 GPUs: 3.5 petaFLOPS per chip; 80% of hyperscaler AI clusters
- Google TPU v5p: Custom tensor processors optimised for Gemini training workloads
- AWS Trainium 2: 4x performance improvement over v1; powers Amazon Nova models
- Intel Gaudi 3: Open ecosystem alternative; 40% cost advantage for inference workloads
- Groq LPU: Ultra-low latency inference chip achieving 500 tokens/second on LLaMA 3
- Apple M4 Neural Engine: On-device AI processing for 2 billion iPhones and Macs

5.3 Edge Computing & On-Device AI

Not all AI runs in the cloud. Edge computing — processing data close to where it is generated — has emerged as a critical architectural pattern for latency-sensitive, privacy-preserving, and bandwidth-constrained applications. The edge AI market reached \$24 billion in 2024 and is growing at 35% annually.

6. Cybersecurity in the AI Era

6.1 AI-Powered Threats

The democratisation of AI has lowered the barrier to entry for malicious actors. 2024 saw a dramatic escalation in AI-enhanced cyberattacks, deepfake fraud, and automated social engineering. Key threat vectors include:

- AI-generated phishing emails with 94% open rates vs 23% for template attacks (Proofpoint, 2024)
- Deepfake voice and video fraud caused \$25.6 billion in financial losses globally
- Automated vulnerability scanning tools reduced attacker reconnaissance time from days to minutes
- LLM-assisted malware generation created 4,000+ novel malware variants in H2 2024
- Nation-state AI campaigns targeted critical infrastructure in 34 countries

Threat Alert: Security researchers identified 'prompt injection' attacks capable of hijacking AI agents to exfiltrate sensitive data, execute unauthorised transactions, or spread disinformation through enterprise AI systems.

6.2 Defensive AI

The same technologies that empower attackers also defend against them. AI-powered security operations centres (AI-SOCs) are now standard at enterprise scale:

- Mean time to detect (MTTD) reduced from 207 days to 26 days with AI threat intelligence
- AI-driven behavioural analytics flagged insider threats with 91% precision
- Automated incident response handled 78% of Tier-1 security alerts without human intervention
- AI model red-teaming identified 340% more vulnerabilities than manual penetration testing

6.3 Zero Trust Architecture

Zero Trust — the principle that no user, device, or system should be trusted by default — has become the de facto security architecture for AI-era organisations. AI enhances zero trust implementations through continuous behavioural risk scoring, adaptive authentication, and real-time policy enforcement across hybrid cloud environments.

7. Emerging Technologies

7.1 Quantum Computing

Quantum computing crossed several commercial viability thresholds in 2024–2025. IBM's 1,386-qubit Heron processor, Google's Willow chip, and IonQ's trapped-ion systems demonstrated fault-tolerant quantum operations on commercially relevant problem sizes. Quantum advantage — the point at which quantum systems outperform classical computers on practical tasks — has been achieved in specific domains:

- Drug molecular simulation: 100x speedup over classical HPC for protein folding problems
- Logistics optimisation: Volkswagen quantum routing reduced fleet fuel costs by 8%
- Cryptography: Post-quantum encryption standards (NIST FIPS 203/204) now mandatory for US federal systems
- Financial modelling: JP Morgan demonstrated quantum Monte Carlo for derivative pricing
- Materials science: Discovery of 3 novel battery cathode materials using quantum simulation

7.2 Autonomous Systems

Autonomous vehicles, drones, and robotic systems reached commercial scale in 2025. Waymo operates 50,000+ autonomous rides per week across multiple US cities. Drone delivery networks serve millions of consumers in Australia, Kenya, and the UAE. Humanoid robotics is entering light manufacturing:

System	Company	Status (2025)	Scale
Robotaxi Service	Waymo	Commercial	50K+ rides/week
Autonomous Trucking	Aurora Innovation	Commercial	TX & PA corridors
Drone Delivery	Wing (Google)	Commercial	5 countries
Humanoid Robot	Figure AI	Pilot	BMW factories
Warehouse Robot	Amazon Sparrow	Deployed	500+ facilities
Autonomous Ships	Kongsberg Maritime	Coastal ops	15 vessels

7.3 Spatial Computing & AR/VR

Apple Vision Pro and Meta Quest 3 established spatial computing as a mainstream category in 2024. Enterprise applications — remote collaboration, surgical training, industrial maintenance — are outpacing consumer entertainment use cases. AI is central to spatial computing, powering scene understanding, hand tracking, and contextual assistance in three-dimensional environments.

7b. AI in Science & Research

Scientific Discovery Acceleration

One of the most profound impacts of AI is its ability to accelerate scientific discovery at a pace impossible for unaided human researchers. AI systems are now co-authoring scientific papers, designing experiments, analysing results, and generating hypotheses across every branch of science:

- **Climate Science:** AI models from Google DeepMind improved 10-day weather forecast accuracy to match 15-day human ensemble models
- **Astronomy:** AI catalogued 1.2 million previously undetected galaxies in ESA Euclid telescope data within 72 hours of release
- **Particle Physics:** Machine learning at CERN's Large Hadron Collider filters 40 million collision events per second to identify physics signals
- **Materials Science:** AI-guided synthesis discovered 381,000 new stable crystal structures, expanding known materials by 45%
- **Genomics:** AI variant-calling tools reduced rare disease diagnosis time from 4.5 years to 6 weeks in clinical trials
- **Mathematics:** Google DeepMind's AlphaProof solved 4 of 6 International Mathematical Olympiad problems at gold-medal level

AI Research Tools & Platforms

Platform	Domain	Key Feature	Users
Semantic Scholar	Literature search	AI-powered relevance ranking	200M+
Elicit	Research synthesis	Automated literature review	4M+
Consensus	Evidence aggregation	AI claim verification	1.5M+
AlphaFold Server	Biology	Free protein structure prediction	2M+ predictions
Wolfram Alpha AI	STEM computation	Symbolic + neural reasoning	50M+
Scite	Citation analysis	Supporting/contrasting citations	1.2M+

Reproducibility & Open Science

AI is also helping address the reproducibility crisis in science. Automated experiment documentation tools, AI-assisted statistical analysis, and open model repositories are making research more transparent and verifiable. The Allen Institute for AI and Hugging Face host over 900,000 open-source AI models and 200,000 datasets, enabling researchers worldwide to build on shared foundations.

Notable 2024 Milestone: AI researchers at Google published the first AI system capable of generating and testing scientific hypotheses autonomously — running 350 computational experiments per day and discovering novel enzyme pathways without human-specified search directions.

7c. AI in Creative Industries

The New Creative Paradigm

The intersection of AI and creativity is redefining what it means to make art, music, games, and films. While debates about authorship and originality continue, the practical reality is that AI tools have dramatically lowered barriers to entry for creative production and expanded the scope of what individual creators can achieve:

Creative Domain	AI Tool Category	Adoption Rate	Impact
Film & Video	AI VFX, deepfake actors, AI scoring	54% of studios	30-60% cost reduction in VFX
Music	AI composition, vocal synthesis, mastering	67% of producers	10x faster demo creation
Game Development	AI NPC dialogue, procedural generation	78% of studios	40% reduction in dev time
Graphic Design	Text-to-image, AI layout tools	82% of designers	5x throughput increase
Fashion	AI trend prediction, virtual try-on	43% of brands	Reduced overstock by 28%
Architecture	Generative design, BIM optimisation	39% of firms	18% material savings
Advertising	AI copy, personalised creative	91% of agencies	34% better conversion rates

Intellectual Property Challenges

The rise of AI-generated content has triggered a wave of legal challenges around copyright, ownership, and training data consent. Key landmark cases in 2024–2025 established important precedents:

- The New York Times v. OpenAI: Settled for undisclosed terms; established precedent for news organisation licensing
- Getty Images v. Stability AI (UK): Court ruled AI training on copyrighted images without licence infringes copyright
- US Copyright Office guidance: AI-generated works without human authorship are not copyrightable
- EU AI Act Article 53: Mandates transparency about copyrighted training data for general-purpose AI models
- Creative Commons AI pledge: 12,000+ artists voluntarily licenced works for ethical AI training datasets

7d. AI Infrastructure & Energy

The Energy Equation

AI's meteoric growth has profound implications for global energy infrastructure. Training a single frontier AI model consumes approximately as much energy as 100 average US homes use in a year. Inference at scale adds further demand. The AI sector's electricity consumption is projected to reach 1,000 TWh annually by 2030 — equivalent to Japan's entire national electricity consumption.

1,000 TWh Projected AI power use by 2030	\$500B Data centre investment planned 2025-2030	40% of new data centres using renewable energy	3x Efficiency improvement from H100 to B200
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Sustainability Responses

The technology industry is responding to AI's energy demands with substantial investments in clean energy and efficiency improvements:

- Microsoft signed 10.5 GW of renewable energy PPAs in 2024 — the largest corporate clean energy deal in history
- Google DeepMind's AlphaControl reduced Google data centre cooling energy by 40% through AI optimisation
- Nvidia's Blackwell architecture delivers 4x the inference performance per watt versus Hopper
- Small modular nuclear reactors (SMRs): Microsoft, Google, and Amazon all signed agreements for nuclear-powered AI campuses
- Liquid cooling adoption reached 62% in new hyperscaler deployments, reducing thermal waste by 35%
- AI model distillation techniques produce 10-100x smaller models with 85-95% of frontier model capability

Data Centre Geography

The distribution of AI compute is becoming a geopolitical issue. Nations are investing in sovereign AI infrastructure to reduce dependency on foreign cloud providers. The EU's GAIA-X initiative, the UK's AI Research Resource, and India's IndiaAI Mission all represent government-sponsored AI infrastructure programmes. Meanwhile, physical constraints — water access for cooling, grid capacity, fibre connectivity — are reshaping where data centres are built, with Nordic countries, Canada, and New Zealand emerging as preferred locations for sustainable AI infrastructure.

8. Ethics, Regulation & Governance

8.1 The EU AI Act

The European Union's AI Act — the world's first comprehensive legislative framework for artificial intelligence — entered into force in August 2024 and began applying progressively through 2025. The Act adopts a risk-based approach, classifying AI systems into four categories:

Risk Level	Examples	Requirements	Penalty (max)
Unacceptable	Social scoring, mass surveillance	Prohibited	N/A
High	CV screening, credit scoring, medical AI	Conformity assessment, registration	€30M / 6% turnover
Limited	Chatbots, deepfake disclosure	Transparency obligations	€15M / 3% turnover
Minimal	Spam filters, AI games	Voluntary codes	None

8.2 Bias, Fairness & Accountability

Algorithmic bias remains a central challenge in AI deployment. Studies in 2024 found measurable disparities across gender, ethnicity, and socioeconomic status in AI systems used for hiring (18% disparity), loan approvals (24% disparity), and criminal risk scoring (31% disparity). Leading organisations are adopting:

- Model cards and datasheets documenting training data, intended use, and known limitations
- Algorithmic impact assessments before high-stakes AI deployment
- Diverse AI development teams — evidence shows teams with 40%+ gender diversity produce 23% fewer biased outputs
- Continuous fairness monitoring with automated alerts when demographic parity drifts

8.3 Responsible AI Frameworks

Constitutional AI, RLHF (Reinforcement Learning from Human Feedback), and interpretability research are enabling AI systems that are safer, more aligned with human values, and more transparent. Anthropic's Constitutional AI, Google's PAIR guidelines, and Microsoft's Responsible AI Standard represent the leading industry frameworks.

9. The Future of Work

9.1 Automation Trends

AI-driven automation is reshaping labour markets at a pace that challenges traditional workforce planning cycles. The World Economic Forum estimates 85 million jobs will be displaced by automation by 2027, while 97 million new roles will be created — but the geographic and skill distribution of these changes creates significant transition challenges.

Role Category	Automation Risk	Timeframe	Mitigation Strategy
Data entry / processing	Very High (87%)	2025–2027	Reskilling to AI oversight
Customer service	High (72%)	2025–2028	Hybrid AI-human models
Legal research	High (68%)	2026–2029	Senior counsel, strategy
Software development	Moderate (45%)	2027–2032	AI-augmented coding
Healthcare diagnosis	Low (18%)	2030+	AI-assisted specialists
Creative direction	Low (12%)	2030+	Human-AI co-creation
Strategic leadership	Very Low (5%)	2035+	Enhanced by AI tools

9.2 Human-AI Collaboration

The most productive organisations are not those that fully automate human roles, but those that design effective human-AI collaboration patterns. Research from MIT Sloan found that human-AI teams outperform either humans or AI alone by 26% on complex analytical tasks, and by 41% on creative problem-solving exercises.

The Centaur Model: Named after the chess concept where human+AI teams beat both grandmasters and supercomputers, the centaur model of work is becoming the standard for high-value professional services in consulting, law, medicine, and engineering.

9.3 Skills of the Future

The half-life of technical skills is shrinking rapidly. Organisations and individuals must embrace continuous learning. The most valued skills in 2025's AI economy are:

- AI literacy: understanding capabilities, limitations, and appropriate use of AI systems
- Prompt engineering and AI interaction design
- Critical thinking and AI output validation
- Ethical reasoning and responsible AI deployment
- Systems thinking: understanding complex sociotechnical systems
- Emotional intelligence: skills that remain uniquely human in collaborative contexts
- Domain expertise combined with AI tool fluency — the 'T-shaped' AI professional

10. Conclusion & Recommendations

10.1 Strategic Takeaways

The technology landscape of 2025 is defined by the convergence of artificial intelligence with every domain of human activity. This is not a temporary disruption — it is a permanent restructuring of how value is created, how work is performed, and how societies organise themselves. Organisations that recognise and act on this reality will lead; those that delay face existential competitive risk.

Six imperatives emerge from this year's research:

- **AI-first strategy:** Treat AI as a core capability, not a technology project. Embed AI thinking into every business function and product decision.
- **Data as strategic asset:** Organisations with high-quality, well-governed proprietary data will build AI moats that are difficult to replicate.
- **Talent transformation:** Invest aggressively in reskilling. The AI talent gap is real but closeable with sustained commitment.
- **Ethical by design:** Build responsible AI practices into development pipelines from day one. Retrofitting ethics is expensive and damaging to trust.
- **Security posture upgrade:** Assume AI-enhanced adversaries. Invest in defensive AI, zero trust, and red team exercises.
- **Regulatory engagement:** Participate constructively in regulatory processes. Early adopters of compliance frameworks will gain competitive advantage.

10.2 Action Plan for Technology Leaders

Horizon	Priority Action	Investment Level	Expected Outcome
0–6 months	Deploy AI productivity tools across org	Low-Medium	20–40% productivity gain
0–6 months	Establish AI governance committee	Low	Risk mitigation framework
6–18 months	Build proprietary data infrastructure	High	Defensible AI advantage
6–18 months	Launch AI reskilling programme	Medium	Workforce readiness
18–36 months	Develop AI-native products/services	High	New revenue streams
18–36 months	Integrate quantum-ready cryptography	Medium	Future-proof security
36+ months	Autonomous AI agent deployment	Very High	Step-change efficiency

"We are at the beginning of the most consequential technological transition in human history. AI is not coming — it is here. The question is not whether to engage, but how to engage wisely."

— Tech & AI Insights, Editorial Board, 2025

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